TEACHER'S COPY — SIDE-BY-SIDE COMPARISON & HSC-STYLE PRACTICE

Arithmetic vs Geometric Sequences

等差数列 与 等比数列 — 对比与练习

Two columns, one page of theory: first the parameter that defines each sequence (d vs r), then the formulas for the nth term and the sum — followed by **5 HSC-style practice questions for APs** and **5 for GPs** (30 marks), with full worked solutions and marking guidelines.

Formulas exactly as printed on the NESA Mathematics Advanced Reference Sheet

AP: $T_n = a + (n-1)d$ • $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + \ell)$, ℓ = last term

GP: $T_n = ar^{n-1}$ • $S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$ $(r \neq 1)$ • $S_\infty = \frac{a}{1-r}$ $(|r| < 1)$

Terms are written T_1, T_2, T_3, … with $a = T_1$ (the sheet prints a_n for the nth term).

1. The defining parameter — common difference d vs common ratio r

Arithmetic sequence (AP) — 等差数列	Geometric sequence (GP) — 等比数列
Definition — how to find it. Subtract consecutive terms: $d = T_2 - T_1 = T_{n+1} - T_n$	**Definition — how to find it. Divide** consecutive terms: $r = \frac{T_2}{T_1} = \frac{T_{n+1}}{T_n}$
What it does each step. Each term = previous term + d. 7, 12, 17, 22, … $\Longrightarrow$ $d = 5$ (add 5 every step)	**What it does each step.** Each term = previous term × r. 7, 21, 63, 189, … $\Longrightarrow$ $r = 3$ (multiply by 3 every step)
Sign and size tell the behaviour. $d > 0 \to$ increasing • $d < 0 \to$ decreasing • $d = 0 \to$ constant ("a, a, a, …")	**Sign and size tell the behaviour.** $r > 1 \to$ growth • $0 < r < 1 \to$ decay • $r < 0 \to$ signs alternate • $r = 1 \to$ constant
Shape of the growth. Linear — the points (n, T_n) lie on a **straight line** (slope d).	**Shape of the growth.** Exponential — the points (n, T_n) lie on a **curve** that multiplies at every step (like compound interest).
Test for three consecutive terms. $T_2 - T_1 = T_3 - T_2 \iff 2T_2 = T_1 + T_3$ The middle term is the **average** of its neighbours.	**Test for three consecutive terms.** $\frac{T_2}{T_1} = \frac{T_3}{T_2} \iff T_2^2 = T_1 T_3$ The middle term **squared** equals the **product** of its neighbours.
From two non-consecutive terms T_m, T_n $(m > n)$: $T_m - T_n = (m-n)d$ — a **linear** equation, so there is always **exactly one** value of d.	**From two non-consecutive terms** T_m, T_n $(m > n)$: $\frac{T_m}{T_n} = r^{m-n}$ — if $m - n$ is **even**, then $r = \pm$ root: **two** values, both valid.
Common errors. Using the term numbers instead of the gap $(m - n)$; losing the sign of d when going backwards.	**Common errors.** Dropping the negative ratio when the power is even; multiplying instead of dividing when stepping backwards.

2. Formulas — nth term and sum of the first n terms

Arithmetic sequence (AP)	Geometric sequence (GP)
nth term (reference sheet): $T_n = a + (n-1)d$	**nth term (reference sheet):** $T_n = ar^{n-1}$
In words. Start at a and **add d** exactly $(n-1)$ times; the index is $n-1$ because the first term has had no step yet.	**In words.** Start at a and **multiply by r** exactly $(n-1)$ times — same reason for the index.
Example. $a = 3$, $d = 4$ — find the 10th term: $T_{10} = 3 + 9(4) = \mathbf{39}$	**Example.** $a = 3$, $r = 4$ — find the 10th term: $T_{10} = 3 \times 4^9 = 3 \times 262\,144 = \mathbf{786\,432}$
Sum of the first n terms (reference sheet): $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + \ell)$, ℓ = last term	**Sum of the first n terms (reference sheet):** $S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$, $r \neq 1$ (if $r = 1$: $S_n = na$)
In words. Number of terms × **average of the first and last terms** — use this form whenever the last term is known.	**In words.** First term × $(1 - r^n)$, divided by $(1 - r)$ — the two forms are the same fraction with top and bottom negated; pick the one that keeps the arithmetic positive.
Example. $a = 3$, $d = 4$, $n = 10$: $S_{10} = \frac{10}{2}(3 + 39) = 5 \times 42 = \mathbf{210}$	**Example.** $a = 3$, $r = 4$, $n = 10$: $S_{10} = \frac{3(4^{10}-1)}{4-1} = 4^{10} - 1 = \mathbf{1\,048\,575}$
"Which term?" / number of terms. Solve a **linear** equation for n: $142 = 7 + (n-1)5 \implies n = 28$	**"Which term?" / number of terms.** Solve an **index** equation for n: $96 = 3 \times 2^{n-1} \implies 2^{n-1} = 32 \implies n - 1 = 5 \implies n = 6$ (or $n - 1 = \log 32 \div \log 2$)
Infinite (limiting) sum. None — an AP eventually grows without bound (only $d = 0$ gives a finite total, namely $S_n = na$).	**Infinite (limiting) sum.** $S_\infty = \frac{a}{1-r}$, **only when $\lvert r \rvert < 1$** (the terms must shrink towards 0).

3. Practice — HSC-style questions (30 marks)

Show all working. Marks for each part are shown in brackets, as in the HSC. Full solutions and marking guidelines are on the next page.

Part A — Arithmetic sequences (15 marks)	Part B — Geometric sequences (15 marks)
1. The 5th term of an arithmetic sequence is 23 and the 12th term is 58. **(a)** Find the first term and the common difference. **(2 marks)** **(b)** Hence find the 20th term. **(1 mark)**	**1.** The 3rd term of a geometric sequence is 12 and the 6th term is 96. **(a)** Find the common ratio and the first term. **(2 marks)** **(b)** Hence find the 8th term. **(1 mark)**
2. Find the sum of the arithmetic series $7 + 12 + 17 + \ldots + 142$. **(3 marks)**	**2.** Find the sum of the first 8 terms of the geometric series $5 + 15 + 45 + \ldots$ **(3 marks)**
3. The first three terms of an arithmetic sequence are $x + 2$, $3x - 1$ and $4x + 1$. Find the value of x and the common difference. **(3 marks)**	**3.** The first three terms of a geometric sequence are $x - 4$, x and $x + 12$. Find the value of x and the common ratio. **(3 marks)**
4. Mia's salary in her first year is \$52 000. Each year afterwards her salary increases by \$1 800. **(a)** Find her salary in the 8th year. **(1 mark)** **(b)** Find her total earnings over the first 8 years. **(2 marks)**	**4.** Liam invests \$4 000 at 6% per annum, compounded annually. **(a)** Explain why the yearly values form a geometric sequence, and state a and r. **(1 mark)** **(b)** Find the value of the investment after 10 years, to the nearest dollar. **(2 marks)**
5. An arithmetic sequence has first term 2, and the sum of its first 10 terms is 155. Find the common difference and the 15th term. **(3 marks)**	**5.** A geometric sequence has first term 18 and limiting sum 27. **(a)** Find the common ratio. **(2 marks)** **(b)** Find the sum of the first 4 terms. **(1 mark)**

4. Worked solutions and marking guidelines

Part A — Arithmetic sequences	Part B — Geometric sequences
1. (a) $T_{12} - T_5 = 7d$: $58 - 23 = 35 = 7d \implies d = 5$ $a = T_5 - 4d = 23 - 20 = 3$ **(b)** $T_{20} = 3 + 19(5) = \mathbf{98}$ $T_{20} = T_{12} + 8d = 58 + 40 = 98$ ✓ Marking: 1 mark d = 5 ▪ 1 mark a = 3 ▪ 1 mark $T_{20} = 98$.	**1. (a)** $\frac{T_6}{T_3} = r^3$: $\frac{96}{12} = 8 = r^3 \implies r = 2$ $a = T_3 \div r^2 = 12 \div 4 = 3$ **(b)** $T_8 = 3 \times 2^7 = 3 \times 128 = \mathbf{384}$ $T_8 = T_6 \times r^2 = 96 \times 4 = 384$ ✓ Marking: 1 mark $r^3 = 8 \to r = 2$ ▪ 1 mark a = 3 ▪ 1 mark $T_8 = 384$.
2. $a = 7$, $d = 5$. Find n from the last term: $142 = 7 + (n-1)5 \implies 135 = 5(n-1) \implies n = 28$ $S_{28} = \frac{28}{2}(7 + 142) = 14 \times 149 = \mathbf{2086}$ Marking: 1 mark forming the equation for n ▪ 1 mark n = 28 ▪ 1 mark $S_{28} = 2086$ (accept $S_n = n/2[2a + (n-1)d]$ fully substituted).	**2.** $a = 5$, $r = \frac{15}{5} = 3$, $n = 8$ $S_8 = \frac{5(3^8 - 1)}{3 - 1} = \frac{5 \times 6560}{2} = \mathbf{16\ 400}$ Marking: 1 mark correct a, r substituted ▪ 1 mark evaluating $3^8 = 6561$ ▪ 1 mark $S_8 = 16\ 400$.
3. Middle = average of neighbours: $2(3x - 1) = (x + 2) + (4x + 1)$ $6x - 2 = 5x + 3 \implies x = 5$ Terms: 7, 14, 21 $\implies d = \mathbf{7}$ Marking: 1 mark forming the equation ▪ 1 mark x = 5 ▪ 1 mark d = 7 (accept listing 7, 14, 21).	**3.** Middle squared = product of neighbours: $x^2 = (x - 4)(x + 12)$ $x^2 = x^2 + 8x - 48 \implies 8x = 48 \implies x = 6$ Terms: 2, 6, 18 $\implies r = \mathbf{3}$ Marking: 1 mark forming the equation ▪ 1 mark x = 6 ▪ 1 mark r = 3 (accept listing 2, 6, 18).
4. (a) Salary is an AP with $a = 52\ 000$, $d = 1800$: $T_8 = 52\ 000 + 7(1800) = \mathbf{\$64\ 600}$ **(b)** $S_8 = \frac{8}{2}(52\ 000 + 64\ 600) = 4 \times 116\ 600 = \mathbf{\$466\ 400}$ Marking: 1 mark $T_8 = \$64\ 600$ ▪ 1 mark correct sum formula with values ▪ 1 mark \$466 400.	**4. (a)** Each year the value is the previous year's value × 1.06 (a 6% increase), so the values form a GP with $a = 4000$ and $r = 1.06$. **(b)** $V_{10} = 4000(1.06)^{10} = 4000 \times 1.790\ 847\ldots = 7163.39\ldots \approx \mathbf{\$7163}$ Marking: 1 mark stating a = 4000, r = 1.06 with reason ▪ 1 mark substituting into $T_n = ar^{n-1}$ with n = 11 or $(1.06)^{10}$ ▪ 1 mark \$7163 (accept 7163.39).
5. $S_{10} = \frac{10}{2}[2(2) + 9d] = 155$ $5(4 + 9d) = 155 \implies 4 + 9d = 31 \implies 9d = 27 \implies d = \mathbf{3}$ $T_{15} = 2 + 14(3) = \mathbf{44}$ Marking: 1 mark substituting into S_n ▪ 1 mark d = 3 ▪ 1 mark $T_{15} = 44$.	**5. (a)** $S_\infty = \frac{a}{1 - r}$: $\frac{18}{1 - r} = 27$ $1 - r = \frac{18}{27} = \frac{2}{3} \implies r = \mathbf{\frac{1}{3}}$ $(|r| < 1$ ✓$)$ **(b)** $S_4 = \frac{18(1 - (⅓)^4)}{1 - ⅓} = \frac{18 \times \frac{80}{81}}{\frac{2}{3}} = \mathbf{\frac{80}{3}} = \mathbf{26\frac{2}{3}}$ Marking: (a) 2 marks for r = ⅓ (1 mark for the equation 18/(1 − r) = 27) ▪ (b) 1 mark $S_4 = 80/3$ (accept 26.67 or 26⅔).

Quick answers — 答案速查. **AP:** 1 (a) $a = 3$, $d = 5$ (b) 98 • 2 2086 • 3 $x = 5$, $d = 7$ • 4 (a) \$64 600 (b) \$466 400 • 5 $d = 3$, $T_{15} = 44$. **GP:** 1 (a) $r = 2$, $a = 3$ (b) 384 • 2 16 400 • 3 $x = 6$, $r = 3$ • 4 (a) $a = 4000$, $r = 1.06$ (b) \$7163 • 5 (a) $r = ⅓$ (b) 80/3 = 26⅔.